

A single-copy lower bound for rank-one nonlinear overlap estimation

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Abstract

Randomized single-copy measurements estimate $\text{Tr}(O\rho^2)$, for $\|O\|_\infty \leq 1$, from $O(\sqrt{d})$ copies of a d -dimensional state at constant accuracy and confidence; this is optimal for observables with large trace norm. We show that low rank does not remove the dimension dependence. For a fixed rank-one projector P , any adaptive single-copy protocol that estimates $\text{Tr}(P\rho^2)$ to error $\varepsilon < 1/8$ with success probability at least $2/3$, for every state, needs more than $(d-1)^{1/3}$ copies when $d \geq 32$. The proof extends partial-transpose telescoping to two ensembles differing only in coherence. For two copies, we find the largest acceptance-probability difference for this pair exactly: $1/(4\sqrt{d-1})$.

Keywords: nonlinear observables, randomized measurements, sample complexity, single-copy measurements, partial transpose, lower bounds

1 Introduction

Motivation

Quantities of the form $\text{Tr}(O\rho^2)$ come up in virtual cooling and in distillation-based error mitigation (see [1] and references therein). Because they are linear in $\rho^{\otimes 2}$, joint measurements on two copies can estimate them, for $\|O\|_\infty \leq 1$, to constant accuracy with a number of copies that does not depend on the dimension d . Many experimental platforms, however, can only measure one copy at a time. In that setting randomized measurements [2] still give estimators, and Du *et al.* [1] showed that well-chosen random measurements estimate $\text{Tr}(O\rho^2)$ for every observable with $\|O\|_\infty \leq 1$ using $O(\max\{\sqrt{d}/\varepsilon, 1/\varepsilon^2\})$ copies, up to a factor $\log[\eta^{-1} \log(\varepsilon^{-1})]$ for failure probability η . For some observables this is known to be optimal. For $O = Z_1$, adaptive single-copy

lower bounds of order $\sqrt{d}$ follow from the hard instances of [3], whose theorem is stated for the normalized quantity $\text{Tr}(Z_1 \rho^2) / \text{Tr}(\rho^2)$ (see the appendix of [1]), and a unitary change of basis extends them to every nonidentity Pauli observable [1]. Bounds of the same order also hold for general observables with $\|O\|_1 = \Omega(d)$, in a restricted range of accuracies [4]. Whether low-rank observables are easier was left open by Du *et al.* [1].

Main result

Low-rank observables, such as projectors onto known reference states, are natural things to measure, so it is reasonable to ask whether knowing such an observable removes the dependence on dimension. We show that it does not, already in the simplest case of a fixed rank-one projector $P = |0\rangle\langle 0|$. Estimating $\text{Tr}(P\rho^2)$ to error $\varepsilon < 1/8$ with success probability $2/3$ by single-copy measurements requires $\Omega(d^{1/3})$ copies (Theorem 3). The bound holds even for the larger class of joint measurements whose effects stay positive under every partial transposition (PPT-BOTH). For the pair of ensembles used in the proof, we also compute the best two-copy advantage exactly (Theorem 4). Finally, we show that the approach used here, summing the individual partially transposed step norms, cannot give an exponent better than $1/3$ (Proposition 14). Closing the gap to the $O(\sqrt{d})$ upper bound of [1] remains an open problem.

Techniques

Our proof uses the partial-transpose telescoping method that Akresh and Beckey introduced for purity and product testing [5]. Adaptive single-copy protocols are first relaxed to PPT-BOTH measurements, and the distinguishing bias is then bounded one copy at a time. The steps coming from the purity-type part of our ensembles are handled as in [5]. What is different here is that our two ensembles differ only in the coherence between $|0\rangle$ and a hidden Haar-random vector. This produces a phase-sensitive step with no counterpart in [5], and computing its exact trace norm (Lemma 11) is the main new calculation.

Related work

Learning-tree arguments give adaptive single-copy lower bounds for purity, inner-product estimation and related tasks [6–8], but none of these concern a fixed rank-one overlap. Lower bounds for state certification with incoherent measurements [9, 10] use random off-diagonal perturbations of a fixed block-diagonal state, and their theorems come with parameter conditions that our pair does not satisfy. For example, [9, Thm. 7.3] assumes the balance condition $d_1\sqrt{a} \leq d_2\sqrt{b}$, which fails for the blocks of our common single-copy average ($d_1 = d - 1$, $d_2 = 1$, $a = 1/(2(d - 1))$, $b = 1/2$) whenever $d \geq 3$. There is also a structural difference: in those results the null hypothesis is a fixed state and the perturbation is small, while in our pair both hypotheses contain a hidden rank-one block tied to the coherence, and the target functional differs by a constant. So these results do not imply Theorem 3. Harrow’s PPT data-hiding bound [11, Thm. 7] needs states that commute with $U^{\otimes N}$ for every $U \in U(d)$, whereas our ensembles are invariant only under the subgroup $1 \oplus U(d - 1)$. Recent protocols for nonlinear properties [12–14] either give upper bounds or use joint operations on

replicas. As far as we know, no earlier result gives a lower bound that grows with the dimension for a fixed rank-one observable in this measurement model.

Organization

Section 2 sets up the model and states the results, and Section 3 introduces the hard pair. Sections 4 and 5 contain the proofs. Section 6 discusses the remaining gap, and Appendix A explains why one natural chain-rule approach fails.

2 Setting and results

Notation

Let $d \geq 3$ and $D := d - 1$. We use the computational basis $|0\rangle, |1\rangle, \dots, |D\rangle$ of $\mathbb{C}^d$. Let $P = |0\rangle\langle 0|$ and $\Pi := I - P$, and identify $\Pi\mathbb{C}^d$ with $\mathbb{C}^D$ through the real isometry $\iota : \mathbb{C}^D \rightarrow \mathbb{C}^d$ that maps the i -th basis vector to $|i\rangle$, $i = 1, \dots, D$. The number of copies is N . We write $D_k := \binom{D+k-1}{k} = \dim \text{Sym}^k(\mathbb{C}^D)$ and $D^{(k)} := D(D+1) \cdots (D+k-1) = k! D_k$, and $\Pi_{\text{sym}}^{(k)}$ for the projector onto $\text{Sym}^k(\mathbb{C}^D) \subset (\mathbb{C}^D)^{\otimes k}$. For a subset S of the copies, Γ_S is the partial transpose of those copies in the computational basis, and $\Gamma_m := \Gamma_{\{m\}}$.

Definition 1 (single-copy protocols) A *single-copy protocol* with N copies measures copy $t = 1, \dots, N$ with a POVM that may depend on all earlier outcomes and on classical randomness, and then outputs a function of the outcomes. No quantum information is carried between copies. For any accept/reject rule applied to its output, the accept effect M on $(\mathbb{C}^d)^{\otimes N}$ is a sum (or integral) of product positive operators $\bigotimes_{t=1}^N F_t$, and so is $I - M$.

Definition 2 (PPT-BOTH [5]) An effect M on $(\mathbb{C}^d)^{\otimes N}$ is *PPT-BOTH* if $0 \leq M^{\Gamma_S} \leq I$ for every subset S of the copies.

Since partial transposition maps product positive operators to product positive operators, the accept effect of any single-copy protocol and its complement stay positive under every Γ_S . In other words, they are PPT-BOTH. For two N -copy states ω, ω' and an effect M , we call $\text{Tr}[M(\omega - \omega')]$ the *advantage* of M . With equal priors, the test that guesses ω when M accepts and ω' otherwise is correct with probability $\frac{1}{2}(1 + \text{advantage})$, so the advantage is twice the improvement over a random guess.

Theorem 3 (main result) Let $d \geq 3$. Suppose a single-copy protocol with N copies outputs, for every state ρ on $\mathbb{C}^d$, an estimate of $\text{Tr}(P\rho^2)$ that is within additive error $\varepsilon < 1/8$ with probability at least $2/3$. Then $N > (d-1)^{1/3}$ whenever $d \geq 32$, and $N \geq (2(d-1))^{1/3}(1-o(1))$ as $d \rightarrow \infty$. The same conclusions hold for any N -copy POVM whose elements remain positive under every partial transposition.

Theorem 3 follows from a bound on how well one specific pair of ensembles can be told apart (Theorem 8). For that pair we can also find the two-copy value exactly.

Theorem 4 (two copies) *For the pair $\omega_c^{(2)}, \omega_m^{(2)}$ of Section 3, the largest advantage $\text{Tr}[M(\omega_c^{(2)} - \omega_m^{(2)})]$ is the same over single-copy protocols and over PPT-BOTH effects, and equals $1/(4\sqrt{d-1})$.*

3 The hard pair and its moment structure

Let u be a Haar-random unit vector in $\Pi\mathbb{C}^d$, drawn once and shared by all copies. Define

$$\rho_c(u) = |v\rangle\langle v|, \quad |v\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |u\rangle), \quad \rho_m(u) = \frac{1}{2}(P + |u\rangle\langle u|),$$

and $\omega_h^{(N)} := \mathbb{E}_u \rho_h(u)^{\otimes N}$ for $h \in \{c, m\}$.

Lemma 5 (basic properties) *(a) $\text{Tr}(P\rho_c(u)^2) = 1/2$ and $\text{Tr}(P\rho_m(u)^2) = 1/4$ for every u . (b) $\omega_c^{(1)} = \omega_m^{(1)} = \sigma := \frac{1}{2}(P + \Pi/D)$. (c) $\rho_m(u)$ is the average of $|v_\theta\rangle\langle v_\theta|$ over $\theta \in [0, 2\pi)$, where $|v_\theta\rangle = (|0\rangle + e^{i\theta}|u\rangle)/\sqrt{2}$.*

Proof (a) ρ_c is pure, so $\text{Tr}(P\rho_c^2) = \langle 0|\rho_c|0\rangle = 1/2$; since $u \perp |0\rangle$, $\rho_m^2 = \rho_m/2$ and $\text{Tr}(P\rho_m^2) = 1/4$. (b) $\mathbb{E}u = 0$ and $\mathbb{E}|u\rangle\langle u| = \Pi/D$. (c) The cross terms $e^{\pm i\theta}$ average to zero. $\square$

Lemma 6 (reduction) *Suppose a single-copy protocol, or any POVM on $(\mathbb{C}^d)^{\otimes N}$, satisfies the estimation hypothesis of Theorem 3. Let M be the accept effect of the rule “output $\geq 3/8$ ”. Then $\text{Tr}[M(\omega_c^{(N)} - \omega_m^{(N)})] \geq 1/3$.*

Proof Fix u . Under $\rho_c(u)^{\otimes N}$ the target is $1/2$, so an estimate within $\varepsilon < 1/8$ exceeds $3/8$ and the rule accepts with probability at least $2/3$. Under $\rho_m(u)^{\otimes N}$ the target is $1/4$, so an estimate within ε is below $3/8$ and the rule accepts with probability at most $1/3$. Since M does not depend on u and $\text{Tr}[M\rho^{\otimes N}]$ is linear in $\rho^{\otimes N}$, averaging over u gives the claim. $\square$

Moment structure

For a subset S of copies let Q_S be the projector onto vectors whose copies in S lie in $\Pi\mathbb{C}^d$ and whose other copies equal $|0\rangle$. For $w_1, \dots, w_k \in \mathbb{C}^D$ define

$$J_k(w_1 \otimes \dots \otimes w_k) := \binom{N}{k}^{-1/2} \sum_{S=\{s_1 < \dots < s_k\}} \left(\iota w_a \text{ on copy } s_a, |0\rangle \text{ elsewhere} \right),$$

extended linearly to $(\mathbb{C}^D)^{\otimes k}$. The summands for different S are orthogonal, so J_k is an isometry, and the ranges of J_k for different k are orthogonal. Let $H_k := J_k \text{Sym}^k(\mathbb{C}^D)$, of dimension D_k , and $V_k := \bigoplus_{|S|=k} \text{ran}(Q_S \Pi_{\text{sym}}^{(S)})$, where $\Pi_{\text{sym}}^{(S)}$ symmetrizes the copies in S ; thus $\dim V_k = \binom{N}{k} D_k$ and $H_k \subset V_k$. We write P_X for the projector onto a subspace X .

Proposition 7

$$\omega_c^{(N)} = 2^{-N} \sum_{k=0}^N \binom{N}{k} \frac{P_{H_k}}{D_k}, \quad \omega_m^{(N)} = 2^{-N} \sum_{k=0}^N \frac{P_{V_k}}{D_k}.$$

Proof Expanding the tensor power, with $|0^{S^c} u^S\rangle$ denoting u on the copies in S and $|0\rangle$ elsewhere,

$$|v\rangle^{\otimes N} = 2^{-N/2} \sum_S |0^{S^c} u^S\rangle = 2^{-N/2} \sum_{k=0}^N \binom{N}{k}^{1/2} J_k(u^{\otimes k}). \quad (1)$$

The substitution $u \mapsto e^{i\theta} u$ preserves the Haar measure and multiplies $J_k(u^{\otimes k}) J_l(u^{\otimes l})^\dagger$ by $e^{i(k-l)\theta}$, so the average of this term vanishes unless $k = l$. We call this the *phase rule* and use it repeatedly. With $\mathbb{E}(|u\rangle\langle u|)^{\otimes k} = \Pi_{\text{sym}}^{(k)}/D_k$ [15] we obtain $\omega_c^{(N)} = 2^{-N} \sum_k \binom{N}{k} J_k \Pi_{\text{sym}}^{(k)} J_k^\dagger / D_k$, and $J_k \Pi_{\text{sym}}^{(k)} J_k^\dagger = P_{H_k}$ because J_k is an isometry. For the mixed ensemble, $\rho_m^{\otimes N} = 2^{-N} \sum_S P^{\otimes S^c} \otimes (|u\rangle\langle u|)^{\otimes S}$; the average of the S -term is $Q_S \Pi_{\text{sym}}^{(S)} / D_{|S|}$, and these projectors are orthogonal for different S . $\square$

For $N = 2$ the blocks $k = 0, 2$ coincide ($H_k = V_k$), $P_{V_1} = \sum_i (|0i\rangle\langle 0i| + |i0\rangle\langle i0|)$ and $P_{H_1} = \frac{1}{2} \sum_i (|0i\rangle\langle 0i| + |i0\rangle\langle i0|)(\langle 0i| + \langle i0|)$. Hence

$$\Delta_2 := \omega_c^{(2)} - \omega_m^{(2)} = \frac{2P_{H_1} - P_{V_1}}{4D} = \frac{Y}{4D}, \quad Y := \sum_{i=1}^D (|i0\rangle\langle 0i| + |0i\rangle\langle i0|). \quad (2)$$

4 Proof of Theorem 3

Theorem 8 (bias bound) *For every $N \geq 2$ and every PPT-BOTH effect M on $(\mathbb{C}^d)^{\otimes N}$,*

$$\begin{aligned} |\text{Tr}[M(\omega_c^{(N)} - \omega_m^{(N)})]| &\leq \beta_N := \sum_{m=2}^N \left(E_m + \frac{1}{2} \|W_m\|_1 \right) \\ &\leq \frac{N(N-1)}{4D} + \frac{1}{2\sqrt{2D}} \sum_{j=1}^{N-1} \sqrt{j} \leq \frac{N(N-1)}{4D} + \frac{N^{3/2}}{3\sqrt{2D}}, \end{aligned}$$

where $E_m := \mathbb{E} K / (D + K)$ and $\|W_m\|_1 = \mathbb{E} \sqrt{(m-1-L)/(D+L)}$ (Lemma 11), with $K, L \sim \text{Bin}(m-1, \frac{1}{2})$.

The proof has four parts. Telescoping (Lemma 9) bounds the advantage by a sum of one-copy steps. Each step splits into a purity-type part and a coherent part (Section 4.2). The purity-type parts are bounded as in [5] (Lemma 10), and the coherent part is computed exactly (Lemma 11). Section 4.5 combines the bounds.

4.1 Telescoping [5]

Lemma 9 *Let $\omega^{(m)}$, $m = 0, \dots, N$, be states on m copies with $\omega^{(0)} = 1$ and $\text{Tr}_m \omega^{(m)} = \omega^{(m-1)}$. Let σ be a state on one copy, and set $X_m := \omega^{(m)} - \omega^{(m-1)} \otimes \sigma$. For every*

PPT-BOTH effect M ,

$$|\text{Tr}[M(\omega^{(N)} - \sigma^{\otimes N})]| \leq \frac{1}{2} \sum_{m=1}^N \|X_m^{\Gamma_m}\|_1.$$

Proof We have $\omega^{(N)} - \sigma^{\otimes N} = \sum_{m=1}^N X_m \otimes \sigma^{\otimes(N-m)}$. Partial transposition preserves the pairing, $\text{Tr}[AB] = \text{Tr}[A^{\Gamma_m} B^{\Gamma_m}]$, and $(X_m \otimes \sigma^{\otimes(N-m)})^{\Gamma_m} = X_m^{\Gamma_m} \otimes \sigma^{\otimes(N-m)}$ because Γ_m acts on copy m only. This operator is Hermitian and traceless and $0 \leq M^{\Gamma_m} \leq I$, so the absolute value of its pairing with M^{Γ_m} is at most half its trace norm, which equals $\|X_m^{\Gamma_m}\|_1$. $\square$

We apply Lemma 9 to $\omega^{(m)} = \omega_c^{(m)}$ and to $\omega^{(m)} = \omega_m^{(m)}$, with the common σ of Lemma 5(b), and combine the two bounds by the triangle inequality. The $m = 1$ terms vanish because $\omega_c^{(1)} = \omega_m^{(1)} = \sigma$. Below we use that P , Π and ι are real, so transposition fixes P and Π , and $(|0\rangle\langle u|)^T = |\bar{u}\rangle\langle 0|$, where $\bar{u}$ is the entrywise complex conjugate of u .

4.2 Splitting a step

Write $c_k := \binom{m-1}{k}$. Expanding the last copy with $\rho_c = \frac{1}{2}(P + |0\rangle\langle u| + |u\rangle\langle 0| + |u\rangle\langle u|)$ and $\sigma = \frac{1}{2}(P + \Pi/D)$, the P -terms cancel, and

$$\begin{aligned} X_m^c &:= \omega_c^{(m)} - \omega_c^{(m-1)} \otimes \sigma = \frac{1}{2}(A_m + B_m), \\ X_m^m &:= \omega_m^{(m)} - \omega_m^{(m-1)} \otimes \sigma = \frac{1}{2}A'_m, \\ A_m &:= \mathbb{E}[\rho_c^{\otimes(m-1)} \otimes |u\rangle\langle u|] - \omega_c^{(m-1)} \otimes \Pi/D, \\ B_m &:= \mathbb{E}[\rho_c^{\otimes(m-1)} \otimes (|0\rangle\langle u| + |u\rangle\langle 0|)], \\ A'_m &:= \mathbb{E}[\rho_m^{\otimes(m-1)} \otimes |u\rangle\langle u|] - \omega_m^{(m-1)} \otimes \Pi/D. \end{aligned}$$

Hence $\|(X_m^c)^{\Gamma_m}\|_1 \leq \frac{1}{2}\|A_m^{\Gamma_m}\|_1 + \frac{1}{2}\|B_m^{\Gamma_m}\|_1$ and $\|(X_m^m)^{\Gamma_m}\|_1 = \frac{1}{2}\|A'_m{}^{\Gamma_m}\|_1$.

4.3 Purity-type steps

Lemma 10 $\|A_m^{\Gamma_m}\|_1 \leq 2E_m$ and $\|A'_m{}^{\Gamma_m}\|_1 \leq 2E_m$. Moreover $E_m \leq (m-1)/(2D)$.

Proof Step 1: the basic operator. For $k \geq 0$ define, on $(\mathbb{C}^D)^{\otimes(k+1)}$ with slots $1, \dots, k+1$,

$$Y_k := (\Pi_{\text{sym}}^{(k+1)}/D_{k+1})^{\Gamma_{k+1}} - (\Pi_{\text{sym}}^{(k)}/D_k) \otimes I_D/D.$$

We show $\|Y_k\|_1 \leq 2k/(D+k)$; this is the purity step of [5] in dimension D . Write $\Pi_{\text{sym}}^{(k+1)}/D_{k+1} = (1/D^{(k+1)}) \sum_{\pi \in S_{k+1}} P_\pi$, where P_π permutes the slots. The permutations fixing slot $k+1$ sum to $k! \Pi_{\text{sym}}^{(k)} \otimes I_D$, are unaffected by Γ_{k+1} , and contribute $(\Pi_{\text{sym}}^{(k)}/D_k) \otimes I_D/(D+k)$. Subtracting the second term of Y_k leaves $-\frac{k}{D(D+k)} (\Pi_{\text{sym}}^{(k)}/D_k) \otimes I_D$, whose trace norm is $k/(D+k)$. Every other permutation factors uniquely as $\pi = \pi' \circ (a \ k+1)$

with π' fixing slot $k+1$ and $a \in \{1, \dots, k\}$. Let $|\Phi\rangle_{a,k+1} := \sum_{i=1}^D |i\rangle_a |i\rangle_{k+1}$ be the unnormalized maximally entangled vector of slots a and $k+1$ (with the identity on the other slots); then $P_{(a,k+1)}^{\Gamma_{k+1}} = |\Phi\rangle\langle\Phi|_{a,k+1}$. Hence the moving permutations contribute $(k!/D^{(k+1)})(\Pi_{\text{sym}}^{(k)} \otimes I_D) G$ with $G := \sum_{a=1}^k |\Phi\rangle\langle\Phi|_{a,k+1}$. Since G is invariant under permutations of slots $1, \dots, k$, it commutes with $\Pi_{\text{sym}}^{(k)} \otimes I_D$, so this contribution is positive semidefinite and its trace norm is its trace. As $\text{Tr}[P_{\pi}^{\Gamma_{k+1}}] = \text{Tr}[P_{\pi}] = D^{\#\text{cycles}(\pi)}$, $\sum_{\pi \in S_{k+1}} D^{\#\text{cycles}(\pi)} = D^{(k+1)}$, and the permutations fixing slot $k+1$ contribute $D \cdot D^{(k)}$ to this sum, the trace is $(D^{(k+1)} - D \cdot D^{(k)})/D^{(k+1)} = k/(D+k)$. The triangle inequality gives $\|Y_k\|_1 \leq 2k/(D+k)$.

Step 2: the coherent ensemble. Apply (1) to $m-1$ copies, with J_k defined for $m-1$ copies. By the phase rule,

$$\mathbb{E}[\rho_c^{\otimes(m-1)} \otimes |u\rangle\langle u|] = 2^{-(m-1)} \sum_k c_k (J_k \otimes \iota) \mathbb{E}(|u\rangle\langle u|)^{\otimes(k+1)} (J_k \otimes \iota)^\dagger,$$

where the last factor of $(|u\rangle\langle u|)^{\otimes(k+1)}$ acts on $\mathbb{C}^D$, and similarly $\omega_c^{(m-1)} \otimes \Pi/D = 2^{-(m-1)} \sum_k c_k (J_k \otimes \iota) (\Pi_{\text{sym}}^{(k)} / D_k) \otimes I_D / D (J_k \otimes \iota)^\dagger$. Because ι is real, $((J_k \otimes \iota) X (J_k \otimes \iota)^\dagger)^{\Gamma_m} = (J_k \otimes \iota) X^{\Gamma_{k+1}} (J_k \otimes \iota)^\dagger$ for every operator X on $(\mathbb{C}^D)^{\otimes(k+1)}$. With $\mathbb{E}(|u\rangle\langle u|)^{\otimes(k+1)} = \Pi_{\text{sym}}^{(k+1)} / D_{k+1}$ this gives

$$A_m^{\Gamma_m} = 2^{-(m-1)} \sum_k c_k (J_k \otimes \iota) Y_k (J_k \otimes \iota)^\dagger.$$

The k -th summand is an isometric image of Y_k supported on $\text{ran } J_k \otimes \Pi \mathbb{C}^d$, and these subspaces are mutually orthogonal. So the coherent k -sector is a single copy of Y_k with weight $2^{-(m-1)} c_k$, and $\|A_m^{\Gamma_m}\|_1 \leq 2^{-(m-1)} \sum_k c_k 2k/(D+k) = 2E_m$.

Step 3: the mixed ensemble. Expanding $\rho_m = \frac{1}{2}(P + |u\rangle\langle u|)$ gives

$$\mathbb{E}[\rho_m^{\otimes(m-1)} \otimes |u\rangle\langle u|] = 2^{-(m-1)} \sum_{S \subseteq \{1, \dots, m-1\}} P^{\otimes S^c} \otimes \mathbb{E}(|u\rangle\langle u|)^{\otimes(S \cup \{m\})},$$

and by Proposition 7, $\omega_m^{(m-1)} \otimes \Pi/D = 2^{-(m-1)} \sum_S (Q_S \Pi_{\text{sym}}^{(S)} / D_{|S|}) \otimes \Pi/D$. For each S with $|S| = k$, the difference of the two S -terms, after Γ_m , is an isometric image of Y_k with weight $2^{-(m-1)}$, supported on $\text{ran } Q_S \otimes \Pi \mathbb{C}^d$; these supports are mutually orthogonal. There are c_k subsets with $|S| = k$. Thus the mixed k -sector consists of c_k orthogonal copies of Y_k , each of weight $2^{-(m-1)}$, while the coherent k -sector is one copy of weight $2^{-(m-1)} c_k$: the multiplicity $\binom{m-1}{k}$ sits in the number of blocks in one case and in the coefficient in the other, as in Proposition 7. The total k -weights agree, and $\|A_m^{\Gamma_m}\|_1 \leq 2E_m$.

Finally, $E_m \leq \mathbb{E}K/D = (m-1)/(2D)$. $\square$

4.4 Coherent steps

Lemma 11 (exact coherent step) *Let $W_m := \mathbb{E}[\rho_c^{\otimes(m-1)} \otimes |\bar{u}\rangle]$, viewed as a linear map from $(\mathbb{C}^d)^{\otimes(m-1)}$ to $(\mathbb{C}^d)^{\otimes(m-1)} \otimes \mathbb{C}^d$. Then $\|B_m^{\Gamma_m}\|_1 = 2\|W_m\|_1$ and*

$$\|W_m\|_1 = 2^{-(m-1)} \sum_{l=0}^{m-2} \sqrt{c_{l+1} c_l \frac{l+1}{D+l}} = \mathbb{E}_{L \sim \text{Bin}(m-1, \frac{1}{2})} \sqrt{\frac{m-1-L}{D+L}} \leq \sqrt{\frac{m-1}{2D}}.$$

Proof (i) Reduction to W_m . Since $(|0\rangle\langle u|)^T = |\bar{u}\rangle\langle 0|$ and $(|u\rangle\langle 0|)^T = |0\rangle\langle \bar{u}|$, we have $B_m^{\Gamma_m} = Z + Z^\dagger$ with $Z := \mathbb{E}[\rho_c^{\otimes(m-1)} \otimes |\bar{u}\rangle\langle 0|] = W_m \otimes \langle 0|$. Let $V_0 := (\mathbb{C}^d)^{\otimes(m-1)} \otimes |0\rangle$ and $V_\perp := (\mathbb{C}^d)^{\otimes(m-1)} \otimes \Pi\mathbb{C}^d$. Then Z maps V_0 into $V_\perp$ and vanishes on $V_\perp$, so $Z + Z^\dagger$ is block off-diagonal with respect to $V_0 \oplus V_\perp$. Its nonzero eigenvalues are $\pm$ the nonzero singular values of Z , and $\|B_m^{\Gamma_m}\|_1 = 2\|Z\|_1 = 2\|W_m\|_1$.

(ii) Block decomposition. Expand $\rho_c^{\otimes(m-1)} = (|v\rangle\langle v|)^{\otimes(m-1)}$ with (1). A term $J_k(u^{\otimes k})J_l(u^{\otimes l})^\dagger \otimes \iota\bar{u}$ acquires the factor $e^{i(k-l-1)\theta}$ under $u \mapsto e^{i\theta}u$, so only $k = l+1$ survives:

$$W_m = 2^{-(m-1)} \sum_{l=0}^{m-2} \sqrt{c_{l+1}c_l} (J_{l+1} \otimes \iota) T_l J_l^\dagger, \quad T_l := \mathbb{E}_u [u^{\otimes(l+1)} \otimes \bar{u} (u^{\otimes l})^\dagger].$$

Here u is Haar-random on the unit sphere of $\mathbb{C}^D$, and T_l maps $(\mathbb{C}^D)^{\otimes l}$ to $(\mathbb{C}^D)^{\otimes(l+1)} \otimes \mathbb{C}^D$. The l -th block vanishes on the orthogonal complement of $\text{ran } J_l$ and takes values in $\text{ran } J_{l+1} \otimes \Pi\mathbb{C}^d$; both families of subspaces are mutually orthogonal in l . Hence $\|W_m\|_1 = 2^{-(m-1)} \sum_l \sqrt{c_{l+1}c_l} \|T_l\|_1$.

(iii) Singular values of T_l . T_l vanishes on the orthogonal complement of $\text{Sym}^l(\mathbb{C}^D)$. Replacing u by Uu , $U \in U(D)$, preserves the Haar measure and gives $T_l U^{\otimes l} = (U^{\otimes(l+1)} \otimes \bar{U}) T_l$. Hence $T_l^\dagger T_l$ commutes with every $U^{\otimes l}$; since $U^{\otimes l}$ acts irreducibly on $\text{Sym}^l(\mathbb{C}^D)$, Schur's lemma gives $T_l^\dagger T_l = \tau_l \Pi_{\text{sym}}^{(l)}$ for some $\tau_l \geq 0$. Taking traces, with u' an independent copy of u and the moment formula of [15],

$$\tau_l D_l = \|T_l\|_F^2 = \mathbb{E}_{u,u'} \langle u, u' \rangle^{l+1} \langle \bar{u}, \bar{u}' \rangle \langle u', u \rangle^l = \mathbb{E} |\langle u, u' \rangle|^{2(l+1)} = 1/D_{l+1}.$$

So T_l has D_l nonzero singular values, all equal to $(D_l D_{l+1})^{-1/2}$, and $\|T_l\|_1 = \sqrt{D_l/D_{l+1}} = \sqrt{(l+1)/(D+l)}$.

(iv) Simplification. Since $c_{l+1}(l+1) = c_l(m-1-l)$, the l -th summand is

$$2^{-(m-1)} \sqrt{c_{l+1}c_l} \frac{l+1}{D+l} = 2^{-(m-1)} c_l \sqrt{\frac{m-1-l}{D+l}},$$

and the term $l = m-1$ of the binomial expectation vanishes. For the bound, use $D+L \geq D$ and Jensen's inequality, $\mathbb{E}\sqrt{m-1-L} \leq \sqrt{(m-1)/2}$. $\square$

4.5 Conclusion of the proofs

Proof of Theorem 8 By Lemma 9, the splitting, and Lemmas 10 and 11,

$$|\text{Tr}[M(\omega_c^{(N)} - \sigma^{\otimes N})]| \leq \frac{1}{2} \sum_{m=2}^N (E_m + \|W_m\|_1), \quad |\text{Tr}[M(\omega_m^{(N)} - \sigma^{\otimes N})]| \leq \frac{1}{2} \sum_{m=2}^N E_m.$$

The triangle inequality gives β_N . The simplified bounds follow from $E_m \leq (m-1)/(2D)$, $\|W_m\|_1 \leq \sqrt{(m-1)/(2D)}$ and $\sum_{j=1}^{N-1} \sqrt{j} \leq \int_0^N \sqrt{x} dx = \frac{2}{3} N^{3/2}$. $\square$

Proof of Theorem 3 Let M be the accept effect of Lemma 6. It is PPT-BOTH: for single-copy protocols by the observation after Definition 2, and for a POVM whose elements remain positive under every partial transposition because M and $I - M$ are sums of such elements. For $N \leq 1$ the two ensembles induce the same N -copy state (trivially for $N = 0$, and by Lemma 5(b) for $N = 1$), so every effect has advantage 0, contradicting Lemma 6; hence $N \geq 2$. Lemma 6 and Theorem 8 give $1/3 \leq \beta_N \leq N(N-1)/(4D) + N^{3/2}/(3\sqrt{2D})$. If $N \leq D^{1/3}$, the right-hand side is at most $D^{-1/3}/4 + 1/(3\sqrt{2})$, which decreases in D and equals $0.31528 \dots <$

$1/3$ at $D = 31$; this contradiction gives $N > D^{1/3}$ for $d \geq 32$. If $N \leq (1 - \delta)(2D)^{1/3}$ for a fixed $\delta \in (0, 1)$, the right-hand side is at most $(1 - \delta)^2(2D)^{2/3}/(4D) + (1 - \delta)^{3/2}/3$, which tends to $(1 - \delta)^{3/2}/3 < 1/3$. Hence $N > (1 - \delta)(2D)^{1/3}$ for all sufficiently large d , for every fixed δ . $\square$

Remark 12 (finite certificates) Solving $\beta_N \geq 1/3$ with the exact sum gives slightly larger finite thresholds than the simplified inequality. For example, at $d = 64$ the exact sum requires $N \geq 6$, while the simplified bound certifies $N \geq 5$. The simplified threshold is asymptotic to $(2(d - 1))^{1/3}$. We do not claim that either threshold is the optimal sample complexity.

5 Proof of Theorem 4

Upper bound

By (2), $(|i0\rangle\langle 0i|)^{\Gamma_2} = |ii\rangle\langle 00|$ and $(|0i\rangle\langle i0|)^{\Gamma_2} = |00\rangle\langle ii|$, so

$$Y^{\Gamma_2} = |\Phi_{\perp}\rangle\langle 00| + |00\rangle\langle \Phi_{\perp}|, \quad |\Phi_{\perp}\rangle := \sum_{i=1}^D |ii\rangle.$$

Since $\langle 00|\Phi_{\perp}\rangle = 0$ and $\|\Phi_{\perp}\| = \sqrt{D}$, the nonzero eigenvalues of Y^{Γ_2} are $\pm\sqrt{D}$, and $\|\Delta_2^{\Gamma_2}\|_1 = 1/(2\sqrt{D})$. For a PPT-BOTH effect, $\text{Tr}[M\Delta_2] = \text{Tr}[M^{\Gamma_2}\Delta_2^{\Gamma_2}] \leq \frac{1}{2}\|\Delta_2^{\Gamma_2}\|_1 = 1/(4\sqrt{D})$, and single-copy protocols are PPT-BOTH.

Achievability

Measure the first copy in the Fourier basis $|f_j\rangle = d^{-1/2} \sum_{k=0}^D e^{2\pi i j k/d} |k\rangle$, $j = 0, \dots, D$, for which $|\langle 0|f_j\rangle|^2 = 1/d$. Given outcome j , let

$$H_j := (\langle f_j| \otimes I) Y (|f_j\rangle \otimes I) = \langle 0|f_j\rangle\langle 0|\Pi f_j| + \langle f_j|0\rangle\langle \Pi f_j|0\rangle,$$

measure the second copy in an eigenbasis of H_j , and accept if the observed eigenvalue is positive. The accept effect is $M = \sum_j |f_j\rangle\langle f_j| \otimes Q_j$, where Q_j projects onto the positive eigenspace of H_j , so by (2) its advantage is $\frac{1}{4D} \sum_j \text{Tr}[Q_j H_j] = \frac{1}{8D} \sum_j \|H_j\|_1$, because each H_j is traceless. As $H_j = |a\rangle\langle b| + |b\rangle\langle a|$ with $a = \langle 0|f_j\rangle|0\rangle \perp b = \Pi f_j$, we get $\|H_j\|_1 = 2\|a\|\|b\| = 2d^{-1/2}(1 - 1/d)^{1/2}$. Summing over the d outcomes gives advantage $\sqrt{D}/(4D) = 1/(4\sqrt{D})$. $\square$

Remark 13 Measuring both copies in the same basis $\{|f_j\rangle\}$ without adaptation gives advantage exactly $1/(2d)$: here $\langle f_k|H_j|f_k\rangle = \frac{2}{d}(\delta_{jk} - \frac{1}{d})$, and the best accept set gives $\frac{1}{8D} \sum_{j,k} |\langle f_k|H_j|f_k\rangle| = \frac{1}{2d}$. We do not claim that this is the best nonadaptive protocol.

6 Discussion

The gap

Theorem 3 gives $\Omega(d^{1/3})$ copies for a fixed rank-one P , while [1] gives $O(\sqrt{d})$ (Figure 1). For this pair, the two-copy result rules out the most direct way of closing the gap. At $N = 2$ the optimal advantage for our pair is $\Theta(1/\sqrt{d})$ (Theorem 4), so no bound of the form $C N(N-1)/d$ holds for this pair, even though a bound of that form is what gives $\Omega(\sqrt{d})$ for purity testing in [5]. A bound $C(N/\sqrt{d} + N^2/d)$ on the advantage of single-copy effects would give $\Omega(\sqrt{d})$, but we do not know whether it holds.

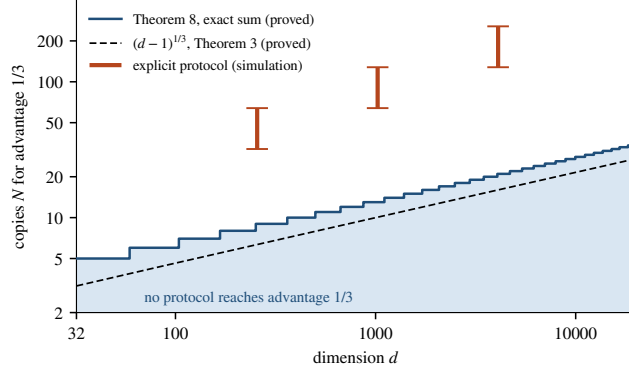


Fig. 1 Copies needed to reach advantage $1/3$ on the pair of Section 3, which any estimator with error $\varepsilon < 1/8$ and success probability $2/3$ must reach (Lemma 6). Below the solid steps (exact sum of Theorem 8) no single-copy or PPT-BOTH protocol reaches it; the dashed line is the bound $N > (d-1)^{1/3}$ of Theorem 3. Bars span $N = 2\sqrt{d}$ to $4\sqrt{d}$: the explicit protocol at the end of this section has measured advantage below $1/3$ at the lower end and above it at the upper end (simulation, not used in any proof).

Why summing step norms stops at $d^{1/3}$

One might hope that computing the full step norms exactly, instead of bounding them, would improve the exponent. It would not, as long as the steps are summed with reference σ : by Proposition 14, for $2 \leq N \leq D$ the coherent steps alone add up to a quantity between constant multiples of $N^{3/2}/\sqrt{D}$, and the remaining terms are nonnegative.

Proposition 14 For $2 \leq m \leq D$,

$$\|(X_m^c)^{\Gamma_m}\|_1 \geq \|W_m\|_1 \geq \frac{\sqrt{m-1}}{2\sqrt{2D}}.$$

Consequently, for $2 \leq N \leq D$,

$$\frac{(N-1)^{3/2}}{6\sqrt{2D}} \leq \frac{1}{2} \sum_{m=2}^N \|(X_m^c)^{\Gamma_m}\|_1 \leq \frac{2N^{3/2}}{3\sqrt{2D}}.$$

Proof Let $R := I^{\otimes(m-1)} \otimes (P - \Pi)$. It is a real diagonal unitary, so conjugation by R commutes with Γ_m . In the notation of the splitting, $RA_mR = A_m$, since the last factor of A_m lies in the Π -block, and $RB_mR = -B_m$. Therefore $B_m^{\Gamma_m} = (X_m^c)^{\Gamma_m} - R(X_m^c)^{\Gamma_m}R$, which has trace norm at most $2\|(X_m^c)^{\Gamma_m}\|_1$, and $\|B_m^{\Gamma_m}\|_1 = 2\|W_m\|_1$ by Lemma 11. For the second inequality, $L \leq m-1 \leq D$ gives $D+L \leq 2D$, and $\sqrt{x} \geq x/\sqrt{m-1}$ on $[0, m-1]$ gives $\mathbb{E}\sqrt{m-1-L} \geq \sqrt{m-1}/2$. The lower bound on the sum follows from $\sum_{j=1}^{N-1} \sqrt{j} \geq \frac{2}{3}(N-1)^{3/2}$. For the upper bound, Lemmas 10 and 11 give $\|(X_m^c)^{\Gamma_m}\|_1 \leq E_m + \|W_m\|_1 \leq 2\sqrt{(m-1)/(2D)}$ when $m \leq D$. $\square$

In particular, for $d \geq 6$ and $2(d-1)^{1/3} + 1 \leq N \leq d-1$ the coherent step-norm sum alone is at least $1/3$, so this route cannot certify more than $2(d-1)^{1/3} + 2$ copies. This limitation applies only to arguments that bound the advantage by the sum of the individual partially transposed step norms (Lemma 9) with reference σ . It says nothing about other reference states, other groupings of copies, cancellations between different steps, or a global PPT certificate. Whether the largest PPT-BOTH advantage for this pair is $O(N/\sqrt{d} + N^2/d)$, which would give $\Omega(\sqrt{d})$, is open.

A failed route

A natural approach through the chain rule would bound how well one more copy can be distinguished, given the history so far, by the norm of the coherent posterior mean $\mathbb{E}[u \mid \text{history}]$. Appendix A shows that this bound is false as stated: the two hypotheses can differ in their posterior second moments even when the posterior mean is zero.

A numerical check

The simulation described here is not used in any proof. Since the upper bound of [1] already implies that this pair can be distinguished with $O(\sqrt{d})$ copies, the pair cannot give a lower bound stronger than $\Omega(\sqrt{d})$. To see how an explicit strategy behaves, we simulated one adaptive protocol at $d \in \{256, 1024, 4096\}$, with 4000 trials per hypothesis (standard errors 0.008–0.011). The protocol measures half of the copies in the basis $\{|f_j\rangle\}$ of Section 5 and sets w to the normalized sum of the observed vectors $\Pi|f_j\rangle$; if that sum vanishes, w is a fixed unit vector of $\Pi\mathbb{C}^d$ (this never happened in the runs). It measures the other half in a basis containing $(|0\rangle \pm |w\rangle)/\sqrt{2}$ and decides by the majority sign, breaking ties with a fair coin (bars in Figure 1). At $N = \sqrt{d}$, $2\sqrt{d}$ and $4\sqrt{d}$ (so $N/\sqrt{d-1} \approx 1, 2, 4$) the advantages are about 0.12–0.13, 0.24–0.26 and 0.41–0.42. At each of these three values the three dimensions agree within 1.4 combined standard errors (the largest discrepancy over the whole grid is 2.3, at $N = \sqrt{d}/4$). At larger N the advantage levels off near $1/2$, the most this sign test can reach. These numbers are consistent with the open bound $C(N/\sqrt{d} + N^2/d)$, but since they only bound the optimal advantage from below, they are not evidence for it.

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- **Consent for publication.** Not applicable.
- **Data availability.** The simulation data shown in Figure 1 and described in Section 6, and the saved outputs of the exact and numerical checks, are openly available at <https://github.com/alex-messerlian/rank-one-overlap-bound>.
- **Materials availability.** Not applicable.
- **Code availability.** The scripts used for the exact and numerical checks are openly available at <https://github.com/alex-messerlian/rank-one-overlap-bound> under the MIT License.
- **Author contributions.** The author initiated and directed the project, proposed the problem from the open problems of [1], revised the text, found errors that were corrected, oversaw all agentic workflows, and personally funded the project. The author takes full responsibility for the work.
- **Use of AI tools.** An AI assistant (Anthropic’s Claude models, 2026 releases, used through the Claude Code command-line interface) derived the results and proofs, wrote the verification code, and drafted the text.

Appendix A A counterexample to a posterior-coherence route

Let $d = 4$. Measure copies 1 and 2 in the basis $\{|x_{\pm}\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}, |2\rangle, |3\rangle\}$ and condition on the history (x_+, x_-) . Its probability is $P_c(x_+, x_-) = 7/96$ under the coherent ensemble and $P_m(x_+, x_-) = 11/96$ under the mixed one. Under $\rho_c(u)$ we have $\langle x_{\pm}|v\rangle = (1 \pm u_1)/2$, where u_1 is the coefficient of $|1\rangle$, so the likelihood $|\langle x_+|v\rangle|^2 |\langle x_-|v\rangle|^2$ is invariant under $u \mapsto -u$ and the coherent posterior mean $\mathbb{E}[u | x_+, x_-]$ vanishes. Nevertheless, the conditional states of copy 3,

$$\bar{\rho}_c = \text{diag}\left(\frac{1}{2}, \frac{13}{70}, \frac{11}{70}, \frac{11}{70}\right), \quad \bar{\rho}_m = \text{diag}\left(\frac{1}{2}, \frac{23}{110}, \frac{8}{55}, \frac{8}{55}\right),$$

differ: the measurement $\{|1\rangle\langle 1|, I - |1\rangle\langle 1|\}$ gives the distributions $p_c = (\frac{13}{70}, \frac{57}{70})$ and $p_m = (\frac{23}{110}, \frac{87}{110})$, with $\chi^2(p_c \| p_m) = 108/32683 > 0$. Hence no bound of the form $\chi^2 \leq C \|\mathbb{E}[u | \text{history}]\|^2$ holds. All values are exact rational Haar averages. A similar separation, where one statistic of the transcript vanishes while another quantity that controls the error terms stays large, appears in certification lower bounds [9, App. D]. We include the example only because it rules out this particular route.

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